

Lecture 15: The Six-Factor Formula and Prompt Kinetics

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Reading Assignment

Lamarsh & Baratta (4th Edition):

- **Chapter 7:** Time-Dependent Reactor Physics.
 - Sections 7.1 – 7.2: Reactor Kinetics and Delayed Neutrons.
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1 Review of the Slab Problem: A Chemical Engineering Perspective

In previous lectures, we approached the 1D slab reactor by immediately assuming a steady-state (critical) system. Today, we are going to look at the same problem through the lens of a chemical engineer analyzing a transient transport process.

Instead of forcing the time derivative to zero, we will retain it and solve the full initial value problem. We will see that the spatial and temporal solutions are deeply linked by a single eigenvalue, and this mathematical link is the exact origin of the "Buckling" concept.

1.1 The Transient One-Group Diffusion Equation

We begin with the time-dependent one-group neutron diffusion equation. Recall that the scalar flux is defined as $\phi(x, t) = n(x, t)v$, where n is the neutron density and v is the average neutron speed. The neutron balance equation is:

$$\frac{1}{v} \frac{\partial \phi}{\partial t} = D \frac{\partial^2 \phi}{\partial x^2} - \Sigma_a \phi + \nu \Sigma_f \phi \quad (1)$$

Grouping the production and absorption terms, we get:

$$\frac{1}{v} \frac{\partial \phi}{\partial t} = D \frac{\partial^2 \phi}{\partial x^2} + (\nu \Sigma_f - \Sigma_a) \phi \quad (2)$$

1.2 Separation of Variables

To solve this linear partial differential equation, we employ separation of variables. We assume the flux can be written as the product of a purely time-dependent function and a purely spatial function:

$$\phi(x, t) = F(x)G(t) \quad (3)$$

Substituting this into our governing equation and dividing through by $D \cdot F(x)G(t)$, we separate the variables:

$$\frac{1}{vD} \frac{1}{G(t)} \frac{dG}{dt} - \frac{\nu\Sigma_f - \Sigma_a}{D} = \frac{1}{F(x)} \frac{d^2F}{dx^2} \quad (4)$$

The left side depends only on t , and the right side depends only on x . For this equality to hold for all x and t , both sides must equal a constant. We will call this separation constant $-B^2$.

$$\frac{1}{F(x)} \frac{d^2F}{dx^2} = -B^2 \quad (5)$$

1.3 The Spatial Solution (Geometric Buckling)

The right side yields our familiar spatial eigenvalue problem:

$$\frac{d^2F}{dx^2} + B^2F(x) = 0 \quad (6)$$

For a bare slab reactor of extrapolated width $\tilde{a}$ (centered at $x = 0$), the boundary conditions require the flux to vanish at the extrapolated boundaries: $F(\pm\tilde{a}/2) = 0$.

The general solution is $F(x) = C_1 \cos(Bx) + C_2 \sin(Bx)$. Symmetry dictates $C_2 = 0$. Applying the boundary condition gives us the fundamental eigenvalue:

$$B^2 = \left(\frac{\pi}{\tilde{a}}\right)^2 \quad (7)$$

Because this eigenvalue B^2 is dictated entirely by the physical dimensions and boundary conditions of the reactor, nuclear engineers call it the **Geometric Buckling** (B_g^2).

1.4 The Temporal Solution (Material Buckling)

Now we return to the left side of our separated equation, plugging in $-B_g^2$ for the separation constant:

$$\frac{1}{vD} \frac{1}{G(t)} \frac{dG}{dt} - \frac{\nu\Sigma_f - \Sigma_a}{D} = -B_g^2 \quad (8)$$

Notice the term $\frac{\nu\Sigma_f - \Sigma_a}{D}$. This term is composed entirely of cross-sections and the diffusion coefficient—it represents the neutronic "recipe" of the core. We define this physical property as the **Material Buckling** (B_m^2):

$$B_m^2 \equiv \frac{\nu\Sigma_f - \Sigma_a}{D} \quad (9)$$

Substituting B_m^2 into our time equation and rearranging, we get a simple first-order ordinary differential equation for $G(t)$:

$$\frac{dG}{dt} = vD(B_m^2 - B_g^2)G(t) \quad (10)$$

The solution is a pure exponential:

$$G(t) = G(0)e^{t/\tau} \quad (11)$$

Where the reactor time constant (or period), τ , emerges directly from the difference between the two bucklings:

$$\tau = \frac{1}{vD(B_m^2 - B_g^2)} \quad (12)$$

1.5 The Nuclear Engineering Philosophy: Why Do It This Way?

From a chemical engineering transport perspective, we just solved a transient problem and found the time constant. But look at the elegantly simple stability criteria that fell out of the math:

- **Supercritical** ($B_m^2 > B_g^2$): The period τ is positive. The flux grows exponentially.
- **Subcritical** ($B_m^2 < B_g^2$): The period τ is negative. The flux decays exponentially.
- **Critical** ($B_m^2 = B_g^2$): The period $\tau \rightarrow \infty$. The time derivative is zero, and the reactor operates at steady state.

Chemical engineers are generally comfortable leaving transport properties and geometry coupled together in dimensionless groups (like the Fourier or Thiele moduli). However, reactor designers faced a unique problem: the people enriching the uranium (chemists/physicists) and the people building the tank (mechanical engineers) needed a clean interface to ensure the system would balance precisely at $\tau \rightarrow \infty$.

By defining B_m^2 (the recipe) and B_g^2 (the geometry) as separate metrics that share the same units ($1/\text{cm}^2$), they decoupled the physics from the engineering.

The core designer adjusts the enrichment to achieve a specific B_m^2 . The mechanical engineer adjusts the tank radius to achieve a specific B_g^2 . As long as those two numbers match, the transient vanishes, and the reactor is critical.

2 The "Bridge": From Buckling to the Six-Factor Formula

As we just saw in our transient slab analysis, the stability of a reactor comes down to a balance between two parts:

1. **The Physics (Material Buckling, B_m^2):** The neutronic "recipe."
2. **The Geometry (Geometric Buckling, B_g^2):** The physical leakage.

Last week, we derived the Four-Factor Formula ($\epsilon p f \eta$) to describe the physics of an infinite medium (k_∞). Now, we need to mathematically connect k_∞ to our new concept of leakage. We do this by defining the **Effective Multiplication Factor**, k_{eff} .

2.1 The Six-Factor Formula

For a finite reactor, we multiply k_∞ by the probability that neutrons do *not* leak out. Since neutrons exist in two distinct energy groups (Fast and Thermal), we add two factors:

$$k_{eff} = k_\infty \cdot P_{FNL} \cdot P_{TNL} \quad (13)$$

Expanding k_∞ :

$$k_{eff} = \epsilon p f \eta P_{FNL} P_{TNL} \quad (14)$$

2.2 Defining the Leakage Factors

Using the diffusion theory we derived:

1. Thermal Non-Leakage Probability (P_{TNL}): This is the probability a thermal neutron is absorbed in the core before diffusing out. From the one-group diffusion equation solution, this relates to the Geometric Buckling (B_g^2) and the Diffusion Length (L^2):

$$P_{TNL} = \frac{1}{1 + L^2 B_g^2} \quad (15)$$

Note: If the reactor is infinite, $B_g^2 \rightarrow 0$, so $P_{TNL} \rightarrow 1$.

2. Fast Non-Leakage Probability (P_{FNL}): Fast neutrons also migrate while slowing down. We define a new parameter, the **Fermi Age** (τ_T). Despite the name "Age," it has units of area (cm^2), just like L^2 . It represents (1/6) of the mean square distance a neutron travels from birth (fission) to thermalization.

$$P_{FNL} = e^{-B_g^2 \tau_T} \quad (16)$$

Approximation: For large reactors (small B_g^2), $e^{-x} \approx 1 - x$ or $1/(1 + x)$.

3 Introduction to Reactor Kinetics

In our opening slab problem, we saw that if a reactor's geometry and materials are not perfectly balanced, the neutron population $n(t)$ changes exponentially with time.

Now that we have formally defined k_{eff} to account for both physics and leakage, we can relate this transient behavior directly to k_{eff} .

3.1 The Reactivity (ρ)

We define **Reactivity** as the fractional departure from criticality:

$$\rho = \frac{k_{eff} - 1}{k_{eff}} = \frac{\Delta k}{k} \quad (17)$$

- $\rho = 0$: Critical ($k = 1$).
- $\rho > 0$: Super-critical.
- $\rho < 0$: Sub-critical.

4 Prompt Kinetics (The "Bomb" Approximation)

Let's assume for a moment that all neutrons are emitted instantly at the moment of fission. These are called **Prompt Neutrons**.

4.1 The Kinetic Equation

The rate of change of the neutron population $n(t)$ is:

$$\frac{dn}{dt} = \text{Production} - \text{Loss}$$

$$\text{Production} = k_{eff} \frac{n}{l}, \quad \text{Loss} = \frac{n}{l}$$

Where l is the **Mean Prompt Neutron Generation Time** (lifetime of a prompt neutron from birth to death).

$$\frac{dn}{dt} = \frac{k_{eff}n}{l} - \frac{n}{l} = \frac{k_{eff} - 1}{l}n \quad (18)$$

4.2 The Anatomy of Prompt Neutron Lifetime (l)

We often treat l as a given constant, but it is deeply tied to the physical properties of the core. The lifetime of a prompt neutron consists of two phases:

1. **Slowing-Down Time (t_s):** The time a 2 MeV fast neutron spends colliding with moderator atoms until it reaches thermal equilibrium (0.025 eV). Because the neutron is moving incredibly fast, this takes only about 10^{-7} seconds.
2. **Thermal Diffusion Time (t_d):** The time the now-slow neutron wanders through the core until it is finally absorbed by fuel, poison, or moderator.

Because $t_d \gg t_s$, the total lifetime is dominated by the diffusion phase ($l \approx t_d$). We can calculate this using the macroscopic absorption cross-section (Σ_a) and the thermal neutron velocity (v_{th}):

$$l \approx t_d = \frac{\text{Mean Free Path}}{\text{Velocity}} = \frac{1}{\Sigma_a v_{th}} \quad (19)$$

This equation dictates the intrinsic speed of different reactor designs.

Reactor Type	Moderator	Typical Lifetime (l)
Fast Reactor	None (Fast Spectrum)	10^{-7} s
Light Water Reactor (PWR/BWR)	Light Water (H_2O)	10^{-4} s
Heavy Water Reactor (CANDU)	Heavy Water (D_2O)	10^{-3} s
Graphite Reactor (AGR/RBMK)	Graphite	10^{-3} s

Table 1: Prompt neutron lifetimes. Notice that reactors with low-absorbing moderators (D_2O , Graphite) have longer neutron lifetimes because the neutrons wander longer before being absorbed.

Insight: A heavy water reactor has a neutron lifetime an order of magnitude slower than a light water reactor. This makes it inherently more "sluggish" and slightly easier to control mechanically during a transient.

4.3 The Exponential Excursion

Solving this differential equation yields an exponential growth curve ($n(t) = n_0 e^{t/T}$). Note that nuclear engineers refer to this time constant as the **Reactor Period (T)**, which is mathematically identical to the τ we derived in our slab problem earlier.

Let's plug in numbers for a thermal reactor.

- Mean Generation Time (l): $\approx 10^{-4}$ seconds.
- Reactivity: Suppose we are Supercritical by just 0.1% ($k_{eff} = 1.001$).

$$T = \frac{10^{-4}}{0.001} = 0.1 \text{ seconds}$$

In just **1 second** ($t = 1$), the power rises by $e^{10} \approx 22,000$.

Conclusion: If all neutrons were prompt, a nuclear reactor would be impossible to control. No human or machine could react fast enough.

5 The Saving Grace: Delayed Neutrons

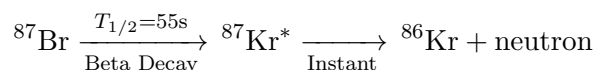
If prompt neutrons were the whole story, nuclear power would not exist. Fortunately, a small fraction of neutrons are **Delayed**.

5.1 The Mechanism: Precursor Decay

When U-235 fissions, it splits into two unstable, neutron-rich fragments. These fragments must undergo Beta Decay ($n \rightarrow p + e^- + \bar{\nu}$) to reach stability. Sometimes, a fragment is so unstable that after the beta decay, the daughter nucleus is highly excited and ejects a neutron directly to reach the ground state.

We call the parent fragment a **Precursor**. The neutron appears "delayed" not because the emission itself is slow, but because it had to wait for the Precursor to undergo beta decay first.

- **Example:** Bromine-87 is a fission product.



5.2 Quantifying the Delay

To understand the impact on the reactor, we look at two parameters: *How many* are delayed (β), and *how long* they wait (τ).

5.2.1 The Delayed Fraction (β)

We define β as the fraction of all fission neutrons that are delayed:

$$\beta = \frac{\text{Delayed Neutrons}}{\text{Total Neutrons}}$$

This value depends heavily on the fuel type, which dictates how "twitchy" (sensitive) a reactor is.

Isotope	Delayed Fraction (β)	Implication
Uranium-235	0.0065 (0.65%)	Standard Control Margin
Plutonium-239	0.0021 (0.21%)	Very Small Margin ("Twitchy")
Uranium-233	0.0026 (0.26%)	Small Margin

Table 2: Delayed Neutron Fractions for common fuels. Note that Pu-239 has $\approx 1/3$ the margin of U-235.

Note that the fuel actually changes during the lifetime of a core. A reactor which starts with LEU (low enriched Uranium) of 5% will have some of its U-238 converted to Pu-239, so the power at the end of core life is produced by fission in both the remaining U-235 and the generated Pu-239. This causes β to decrease over time.

5.2.2 The Effective Time Constant

While ^{87}Br (55s) is the longest-lived precursor, there are dozens of others with half-lives ranging down to **0.2 seconds**.

- **Average Lifetime:** For U-235, the weighted average mean life of a delayed neutron is ≈ 12.7 seconds.
- *Note:* In the next lecture, we will break these down into 6 specific groups to derive the exact control equations.

5.3 The Consequence: Why Delay Creates Stability

The 12.7-second delay acts as a "Time Buffer" that allows both human operators and physical laws to control the reactor.

5.3.1 1. Mechanical Control (The Human Factor)

If the reactor period were determined by prompt neutrons (10^{-4} s), no machine could react fast enough to stop an excursion. The delayed neutrons slow the dominant time constant to seconds or minutes, allowing mechanical control rods to move and human operators to react.

5.3.2 2. Thermal Feedback (The Physics Factor)

Modern reactors rely on negative temperature coefficients for inherent safety, but these feedbacks operate on two completely different time clocks.

- **The Prompt Feedback (Doppler Broadening):** As power spikes, the temperature of the U-238 fuel pellets rises instantaneously because the fission energy is deposited directly within them. This immediately increases resonance absorption of neutrons. Doppler broadening is the reactor's "first line of defense" and blunts the initial power spike instantly without waiting for heat transfer.
- **The Delayed Feedback (Moderator Density):** For the reactor to stabilize, the coolant (water) must heat up, expand, and become less dense, which reduces its ability to thermalize neutrons. However, this heat must physically conduct from the fuel center, across the gap, through the zircaloy cladding, and into the water.

The Race Against Time: Heat transfer to the moderator takes *seconds*. Prompt excursions take *milliseconds*.

- **Without Delayed Neutrons:** If a reactor went prompt critical, the instantaneous Doppler feedback would try to fight the excursion, but the exponential power rise would be so aggressive that the fuel would melt and vaporize before the heat could ever reach the water to trigger the main moderator feedback.
- **With Delayed Neutrons:** The delayed neutrons slow the power rise down to a scale of seconds. This gives the heat enough time to flow through the cladding and into the water, allowing the massive negative feedback of the moderator to kick in and stabilize the reactor safely.

5.4 Defining Criticality Regimes

Because of this distinction, we define two very different states of "Criticality":

- **Delayed Critical ($k = 1$):** Steady state. The chain reaction relies on the delayed neutrons to keep going. Time scale = Seconds/Minutes. (Safe).
- **Prompt Critical ($k \geq 1 + \beta$):** The chain reaction sustains itself on prompt neutrons alone. The delayed neutrons are just "icing on the cake." Time scale = Milliseconds. (Explosive).

6 Case Study: **The Demon Core** (1946)

The most famous example of Prompt Kinetics in action is the accident involving Louis Slotin at Los Alamos. This incident marks the transition from the "cowboy era" of nuclear physics to the modern era of remote safety.

6.1 The Context and Fermi's Warning

Slotin was the lead armorer at Los Alamos and was famous for his bravado in performing an experiment dubbed "Tickling the Dragon's Tail," where he would bring a bare sphere of Plutonium near criticality by hand to verify its readiness. Enrico Fermi famously warned Slotin and his team about this specific manual technique:

"If you keep doing that experiment, you will be dead in a year."

Fermi was tragically correct.

6.2 The Experiment (May 21, 1946)

- **The Core:** A 6.2 kg sphere of Plutonium-239 (The "Rufus" Core, later dubbed the Demon Core).
- **The Reflector:** Two half-spheres of Beryllium (a neutron reflector).
- **The Action:** Slotin held the top half of the reflector with his left hand and used a flathead screwdriver in his right hand to maintain a tiny gap, preventing the reflector from closing completely.

6.3 The Excursion

The screwdriver slipped. The upper Beryllium shell fell onto the core.

1. **Reflection:** The gap closed. Neutrons that were previously leaking ($P_{NL} < 1$) were suddenly reflected back.
2. **Prompt Criticality:** The system went **Prompt Supercritical** ($k_{eff} > 1 + \beta$).
3. **The Blue Flash:** Ionization of the air (and Cherenkov radiation in the eye fluid) caused a bright blue flash.
4. **The Reaction:** Slotin instinctively jerked his hand up, flipping the tamper off. This reaction took a fraction of a second.

6.4 The Result

- **Biological:** Even though the excursion was terminated by thermal expansion and Slotin's reaction ($\sim$ milliseconds), the prompt flux was so high that Slotin received ~ 21 Sv (2100 rem) of neutron/gamma radiation. He died 9 days later from acute radiation poisoning.
- **Material:** The burst produced $\sim 3 \times 10^{15}$ fissions. This created a massive inventory of fission products within the Plutonium sphere itself. The core became too radioactive ("hot") to be used as a weapon core for the upcoming Operation Crossroads tests. It had to be melted down and reprocessed to remove the contaminants.

6.5 The Protocol Shift

This accident ended the era of "hands-on" criticality experiments.

- **Immediate Ban:** Los Alamos immediately banned all manual manipulation of critical assemblies.
- **Remote Assembly:** All future criticality experiments were performed using remote-controlled machines (such as the "Lady Godiva" device) in shielded concrete bunkers (Kivas), with personnel located a quarter-mile away.

Lesson: In a prompt critical system, the "Period" T is shorter than human reaction time. Engineering controls (geometry, remote operation) must replace administrative controls (skill/procedure).

7 Summary

1. k_{eff} combines material properties (k_{∞}) and geometry (P_{NL}).
2. If $k_{eff} > 1$, power rises exponentially.
3. Based on prompt neutrons alone ($l \approx 10^{-4}$ s), the time scale is too fast to control.
4. **Next Lecture:** We will derive the "Inhour Equation" to see exactly how delayed neutrons (β) make nuclear power possible.